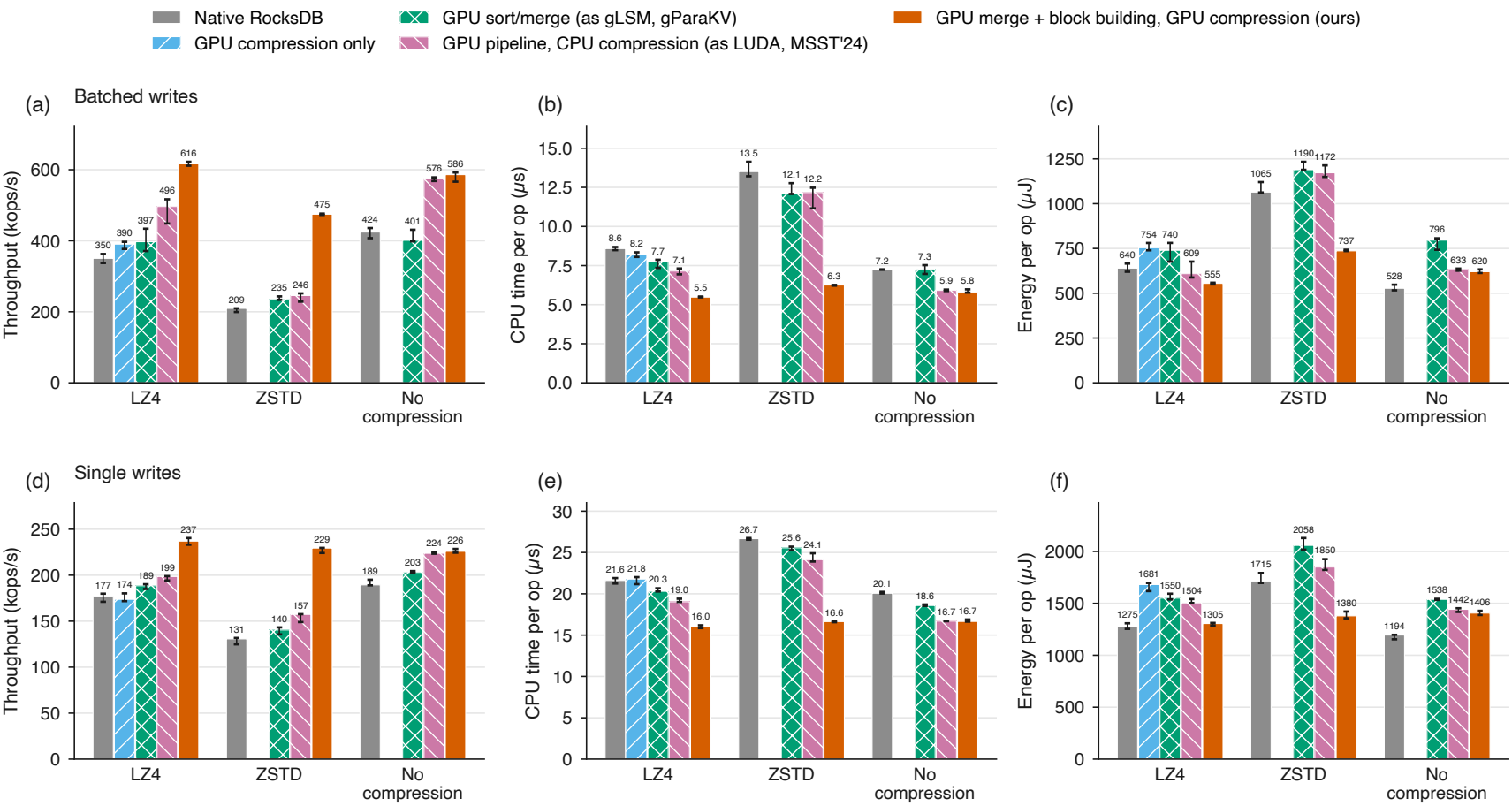
Native RocksDB
GPU compression only
GPU sort/merge (as gLSM, gParaKV)
GPU pipeline, CPU compression (as LUDA, MSST'24)
GPU merge + block building, GPU compression (ours)
(a)
Batched writes
Throughput (kops/s)
350 390 397 496 616
209 235 246 475
424 401 576 586
(b)
CPU time per op (μs)
8.6 8.2 7.7 7.1 5.5
13.5 12.1 12.2 6.3
7.2 7.3 5.9 5.8
(c)
Energy per op (μJ)
640 754 740 609 555
1065 1190 1172 737
528 796 633 620
(d)
Single writes
Throughput (kops/s)
177 174 189 199 237
131 140 157 229
189 203 224 226
(e)
CPU time per op (μs)
21.6 21.8 20.3 19.0 16.0
26.7 25.6 24.1 16.6
20.1 18.6 16.7 16.7
(f)
Energy per op (μJ)
1275 1681 1550 1504 1305
1715 2058 1850 1380
1194 1538 1442 1406
LZ4
ZSTD
No compression